

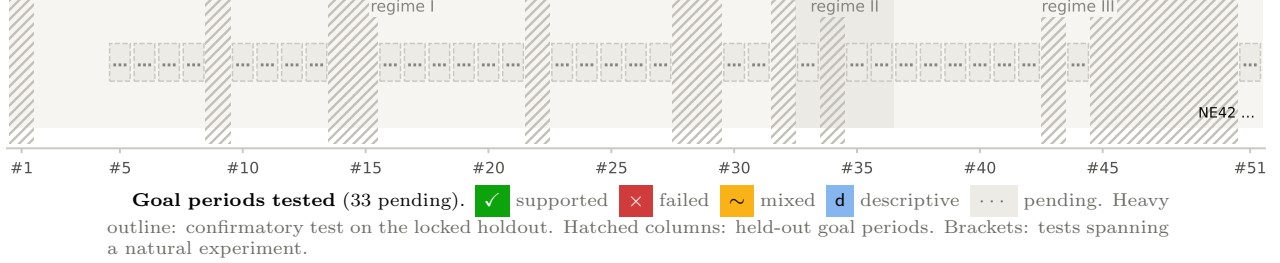
# H41: Read-out gating gives the swarm a light cone

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 2/5 Usefulness 2/5

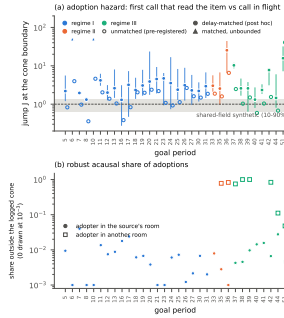
**Bottom line.** Novel items obey a logged light cone inside rooms and across the NE42 merge and split, but the pre-registered gating test mostly failed (it passes only after a post-hoc recency match), and cross-room leaks cannot be traced to any logged channel.

level: hypothesis · pipeline checks 57% · updated 2026-10-04



## WHAT HAPPENED

Can a novel item (a link, number, name or rare word) reach an agent before that agent's model calls could have read it? We traced 167k marker items (H34's rule) through the DQ1 context ledger in 32 non-holdout periods, building each item's logged light cone: the time-respecting read-out paths from its first use. Within a room the bound holds: adoptions outside the cone are rare (robust share 0–2.4% per period). The pre-registered test of gating, the hazard at the first call that read the item over the call already in flight, passed in only 9/32 periods. Matched for time since the item appeared (post hoc), it passes in 21/32. Across rooms the logged cone is a cage: 64–100% of cross-room adoptions lie outside it, and NE42's merge raised the cross-group hazard 80–860×, falling back at the split. The front is paced by talk turns, not cadence. Holdout untouched.



**Observables.** (a) Jump  $J$  per period: open circles are the pre-registered unmatched estimate (often  $\leq 1$  because adoption hazard decays with time since  $t_0$ , and in-flight calls all sit near  $t_0$ ); filled are delay-matched (post hoc). Gray: shared-field synthetic. (b) Share of adoptions outside the logged cone: adopters in the source's room (dots,  $\leq 2.4\%$ ) versus another room (squares, 0.05–1).

## PHYSICAL MODEL

Asynchronous kinetic updates (model 02) with contagion on a temporal graph (03). Agents update only at their model calls; a call reads the room messages posted since the previous call (the ledger). An item posted by source  $s$  at  $t_0$  can reach agent  $i$  only along a time-respecting read-out path;  $K_i$  is  $i$ 's entry call into this logged cone. Adoption hazard per talk call: a shared field  $\varepsilon$ , unlogged channels  $\eta$ , plus  $q$  once  $i$  has read the item.  $J$  compares the hazard at the first entry call with the call in flight at  $t_0$ .

$$T_s = t_0, \quad T_i = \min\{t_c^{\text{call}} : c \text{ reads } m, t_m \geq T_{\text{src}(m)}\}$$

$$\lambda_i(c) = \varepsilon(t_c) + \eta_i(c) + q\mathbb{I}[c \geq K_i^{\text{item}}]$$

$$\text{acausal: } c_{\text{use}} < K_i, \quad J = \frac{h(c = K_i)}{h(c = K_i - 1)} \Big|_{\text{same } t-t_0}$$

$$v = 1/\text{median}(\text{calls from exposure to use})$$

## SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator the useful part is negative: you cannot trace how cross-room information leaked from chat logs. Nearly all cross-room adoptions bypass the logged chat channel, yet shared repos and websites are as common among in-cone adoptions as among the leaks, so they explain nothing. Only humans re-posting an item in several rooms stand out (24× enriched). Rooms work as cages for chat: merging two groups multiplies cross-group pickup about a hundredfold. Inside a room, gating is real but weak: about a third of fast adoptions are co-generated (a shared field), not read. Spread runs at one hop per 1.5–5 talk turns, so tuning call cadence buys nothing.

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## RESULTS VS PREDICTIONS

| Prediction                                                       | Observed                                                             | Verdict              |
|------------------------------------------------------------------|----------------------------------------------------------------------|----------------------|
| P1 $J_{\text{in}}$ CI $> 1$ ( $\geq 70\%$ II/III, $\geq 50\%$ I) | 3/10 II/III, 6/19 I; matched (post hoc) 6/10, 15/19                  | fail (post hoc pass) |
| P2 $A_{\text{rob}} \leq 0.05$ (II/III), 0.10 (I)                 | 29/32; the 3 misses are cross-room; within-room max 0.024            | mostly               |
| P3 early $A$ below time-shuffled null                            | 32/32, but a field also passes (synthetic)                           | non-specific         |
| P4 cross-room adopters unreachable on pre-item graph             | $h = \infty$ for every cross-room adopter; $2 \leq h < \infty$ never | yes                  |
| P5 cycles per hop: I $\leq 5$ , I 47 / 5 talk; III 16 / 1.5 talk | III yes, I no                                                        |                      |
| P6 cadence $b \in [0.5, 1.5]$ , $c > -0.2$                       | $b > 0$ (CI) in 3/25; $c$ often $< 0$ or $> 0$                       | fail                 |
| P7 $\geq 50\%$ of violations have a channel                      | 13/16, but controls 0.84                                             | non-specific         |
| P8 in-flight $\leq$ entry/3                                      | 2/10 (matched 4/10)                                                  | fail                 |
| NE42 merge $\geq 3 \times$ both ways; cage                       | $\times 82$ , $\times 860$ ; cross acausal 1.0 / 0.02 / 1.0          | yes                  |
| G38 cage, artifact leak                                          | cage yes (ratio 0.001); artifact/web 0.19 vs 0.31                    | mixed                |

## GOAL PERIODS IN DETAIL

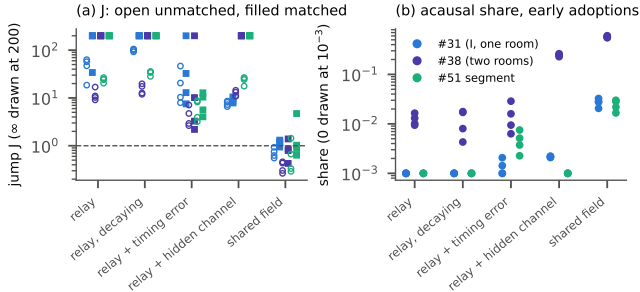
G05 ... pending  
G06 ... pending  
G07 ... pending  
G08 ... pending  
G10 ... pending  
G11 ... pending  
G12 ... pending  
G13 ... pending  
G16 ... pending  
G17 ... pending  
G18 ... pending  
G19 ... pending  
G20 ... pending  
G21 ... pending

+19 more periods: see the card.

## FAITHFULNESS SCORECARD

|                        |   |                   |   |
|------------------------|---|-------------------|---|
| A mapping              | – | F identifiability | – |
| B assumptions          | – | G ground truth    | – |
| C adequacy             | – | H comparative     | – |
| D unfitted predictions | – | I transfer        | – |
| E interventional       | – |                   |   |

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).



**Second figure.** Synthetic validation on real call schedules (#31, #38, #51; 4 replicates of 400 items). (a) The matched  $J$  separates relay ( $\infty$ ) from a shared field ( $\approx 1$ ); timing error attenuates it but keeps the sign. (b) The acausal share separates field from relay only where logged paths are long (two rooms).

## CAVEATS

The headline gating result rests on a post-hoc estimator, chosen after the first real period and validated afterwards (8% false positives under a field). Items are hashed markers, so co-generated names and numbers look like adoptions. Channel attribution is non-specific because shared work is ubiquitous. Visibility is the ledger's room rule; its timing is estimated for non-Gemini agents. The cadence test lacks leverage: regime-III call intervals barely vary. NE42 is confounded with #40's shared goal. The stored per-adoption hop counts were invalid (fixed and recomputed for #51 only). No multiplicity correction across 32 periods. Holdout not run.

## ROUND 2

Pre-register the matched jump for the holdout. Time-stamp the unlogged channel with file-level git reads (DQ4) and web visits against matched controls. Test HH187's cycle-quantized lags on hop events. Use pause timers as exogenous cadence shifts.

*No round-2 redirects yet.*

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